



Q1. I need details of 8051 IC. I have the tool kit for ICs and would appreciate any help from you.

T. Dinesh Kumar
Through e-mail

A1. The Intel 8051 is an 8-bit microcontroller that belongs to the Intel MCS-51 family, commonly referred to as 8051—a microcontroller series developed by Intel in 1980 for use in embedded systems. Intel's original versions were popular in the 1980s and early 1990s, and enhanced derivatives remain popular even today. MCS-51 family developed this using CMOS technology, identified by a letter C in their name (i.e., 80C51 consumes less power, which makes it more suitable for battery-powered applications). Intel no longer manufactures MCS-51, MCS-151 and MCS-251 family chips. While most microcontrollers are made by only one company or two companies, chips in 8051 family are made by dozens of companies with different variants. Current vendors of MCS-51 compatible processors include more than 20 independent manufacturers including Atmel, Infineon Technologies (formerly Siemens AG), Maxim, NXP (formerly Philips Semiconductor) and ST Microelectronics.

Intel 8051 is a very popular microcontroller. Large amounts of information, application notes and tutorials relating to it are available on the Internet. Some of the features that have made 8051 popular are:

- 40-pin device, available in DIP package
- 8-bit ALU and accumulator
- 8-bit registers
- Four fast switchable register banks, with eight registers each
- 4kB on-chip program memory
- 128 bytes on-chip data memory (RAM)

- 16-bit timers (usually two, but may have more, or less)
- Three internal and two external interrupts
- Bit, as well as byte, addressable RAM
- Four 8-bit bi-directional input/output ports
- UART (serial port)

New derivatives with special features are still developed by many major chipmakers—release updates are available on their websites.

The 8051 Microcontroller can be programmed in Assembly, C and a number of other high-level languages. Many compilers even have support for compiling C++ for an 8051. There are various high-level language compilers for 8051 or enhanced versions of 8051 from various vendors. Major compiler suppliers are: Keil (trial version limits code size to 2kB), Raisonance, IAR Systems, mikroC by Mikroelektronika, Tasking, Crossware, SPJ Embedded Technologies India, and SDCC Small Device C Compiler—a completely-free GPL.

USB based programmers—compatible with Windows XP, Windows Vista and Windows 7—are available from different vendors at affordable prices. USB based programmer is a must as new computers and almost all laptops do not have serial or parallel ports.

Select the programmer that supports the entire Atmel AT89 family of microcontrollers. You can make your own development board as a number of circuits are available on the Internet—select a circuit and modify it to suit your requirements.

Q2. How many 1W LEDs can be safely connected to one terminal (one pin output) of IC 4017?

Mohd Waseem Ansari
Through e-mail

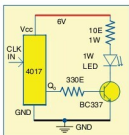
A2. IC 4017 is a 5-stage divide-by-10 Johnson counter.

Ten decoded outputs are normally in logical '0' state and go to logical '1' state only at their respective time slots.

Each decoded output remains high

for one full clock-cycle. The carry-out signal completes a full cycle for every 10-clock input cycles and is used as a ripple-carrying signal to any succeeding stages. 1W white LED requires 350mA through it and 3.2V across it to work. IC 4017 is a low-power device and is not capable to source the current required by the LED.

To overcome this limitation, we have to incorporate individual transistor buffers across the ten



outputs from the IC. Such an arrangement is shown for one output Q_1 in the figure. You can use BC337 npn transistor for the buffer as its maximum collector current rating is 0.8A.

For switching the transistor, connect the bases of the ten individual transistors to the respective outputs of IC 4017 via base resistors. With $I_C = 0.35A$, $h_{FE} = 60$ and overdrive factor = 5, value of the base resistor to drive the transistor into saturation is 200Ω. You can take 220Ω or 330Ω resistors. The LEDs have their own series-limiting resistors.

The LED series dropper resistors' values are calculated using the following relationship:

$R = (V_{CC} - V_{LED}) / I$, where V_{LED} is the voltage drop across LED and I is the optimum current for the LED as specified in its datasheet.

For $V_{CC} = 6V$, $V_{LED} = 3.2V$ and $I = 0.35A$ (typical values), value of $R = 8\Omega$. You can take the nearest value of 10Ω/1W.

Now, your circuit is ready and should be capable of handling any type of high-wattage LED in the range.

Answers compiled by EFY joint director (training), Col. N.C. Pande (Retd). Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender